

Thesis Executive Summary – Progress Update (23 September 2026)

During the past two weeks, I focused mainly on preparing the technical environment and identifying a suitable dataset for the initial Context Grounding prototype.

I selected Synthea synthetic patient data for the initial experiments because it is publicly available and does not require access to sensitive patient data or special permissions. I started examining its clinical data structure, including patient demographics, conditions, encounters, observations, procedures, allergies, and medications. The data preparation and preprocessing are still in progress.

I also established a workflow between Google Colab and GitHub and selected Chroma as the initial vector database.

In addition, I created an initial mind map based on the Context Grounding framework and the broader hallucination-reduction directions previously discussed. The mind map is still at an early stage and will be expanded as the literature review progresses.

My academic literature review is still preliminary, as I spent most of this period on the environment, dataset, and initial prototype setup. My next step is to continue the literature review and start building the first Level 0 Context Grounding prototype.

GitHub Repository: <https://github.com/esrakirbas/msc-thesis-reducing-hallucinations-in-medical-diagnosis>